ENSC-488: Introduction to Robotics

Simon Fraser University

Demo 1 (in ASB 9819)

Lab Project Demo 1 Instructions

Materials to be submitted: You need to hand in the following (printouts/drawings) at the demo time:

As an organized report with Sub-Headings

- A sketch of the robot with all the frames clearly shown.
- Homogenous matrix transformations (symbolic form) for all the link frames ${}^{i-1}_{i}\boldsymbol{T}$, for $i = 1, \ldots, 5$ and station frame ${}^{0}_{S}\boldsymbol{T}$
  - **Note:** frames {0} and {5} are the base frame and tool frame, respectively.
- Symbolic forward kinematic equations for the position (x, y, z) and orientation φ of the tool frame with respect to the station frame.
- All possible inverse kinematic solutions (in symbolic form) as functions of the tool position and orientation, i.e. (x, y, z, φ), with respect to the station frame.
  - Also, specify/list any degenerate/singular cases for inverse kinematic solutions.
- State each teammate's role in the project under the following categories:
  a) Theory/analytical development
  b) Coding and Debugging
  c) Any other category that may be relevant.
- Please do not share code with any other team or reuse code from any team. Teams may share insights freely with each other, even explaining to friends how to solve a particular problem. If any project submission is found to share code/results (of any amount), the entire demo will be graded as a zero for all teams involved.
  a) If using a function/code that is not authored by you, please correctly indicate the function, depth of reuse, and cite its location at the end of the report using IEEE format.

As a packaged zip file

- Submit a copy of all source code used for Demo 1 (i.e. zip file). Ideally, the main.cpp file.
- DO NOT copy code into the report!

Demonstrations:

Your program should demonstrate the following:

- **Emulator:** Move the robot to a given joint configuration of the robot.
  - Joint Values provided by TA/Instructor during the demo.
- **Forward Kinematics**: For the above configuration, compute and report back the corresponding pose (position and orientation) of the tool, i.e., (x,y,z,φ).
- **Inverse Kinematics:** For a given pose of the tool (provided by TA/Instructor), with respect to the station frame, compute and report back all possible robot configurations (joint values).
  - Among these configurations, *indicate rejected ones* that violate joint limits.
  - **Choose** the one "closest" to the current robot configuration. Move the robot to this configuration.
- **Real Robot:** Move the real robot to the configuration computed above, and grasp a real object provided by us in two steps, as follows: first move to a configuration (let us call it **q1**)where end-effector is above (say, by few tens of mm) the object to be grasped, and then lower the gripper to grasp the object. Having grasped the object, move it back to configuration \textbf{q1}, and then to a second configuration **q2** (corresponding to a given placement pose for end-effector, provided by us) so that the gripper is directly above (by few tens of mm) the desired placement pose , and then lower the gripper and release the object on to the table at the placement pose. This essentially completes a simple pick and place task.

Note: Since part of the demo is on the real robot, you will need to use the 488-lab computers for it. Please make sure you have compiled and tested your code on those machines well in advance of the demo date to avoid last minute compilation/version issues. The instructor/TA must grant access to this room supervised.